

UCSC Practical Take-Home Assignment

QA Automation

OrangeHRM Open Source Demo

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Website: <https://opensource-demo.orangehrmlive.com/web/index.php/auth/login>
GitHub: <https://github.com/SeniruR/QA-Automation-ASQA>
Stack: Java 17, Selenium WebDriver 4, TestNG, Maven, Page Object Model

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1 Step 1 - Selected Website (5 Marks)

Item	Detail
Application	OrangeHRM OS Demo (v5.x)
URL	https://opensource-demo.orangehrmlive.com/web/index.php/auth/login
Approval	Lecturer approved
Why this site	Public HRMS demo with Login, Dashboard, PIM, Admin, Leave modules

Demo credentials: Username `Admin` / Password `admin123`

2 Step 2 - Requirement Analysis (10 Marks)

2.1 Application Objectives

OrangeHRM is an open-source Human Resource Management System used to manage employees, leave, time, and system users. The demo site allows exploration of HR workflows through a web UI.

2.2 Target Users

- **System Admin** - manage users, roles, job titles, organisation structure
- **HR / PIM user** - add, search, and edit employee records
- **Employee** - view personal info and apply leave (role-dependent)

2.3 Key Features Observed

1. Authentication (login / logout)
2. Dashboard with widgets and side navigation
3. PIM - employee list, add employee, search filters
4. Admin - system users, job, organisation
5. Leave - apply, assign, list leave
6. Additional modules: Time, Recruitment, My Info, Performance, Directory

2.4 Assumptions

- Demo site remains publicly available during the assignment window
- Default Admin credentials remain `Admin` / `admin123`
- Microsoft Edge or Google Chrome available for automation
- Shared demo data may be reset or changed by other users

2.5 Risks and Mitigation

Risk	Impact	Mitigation
Shared demo data changes	Flaky search results	Prefer stable flows; read-only searches
Slow page load	Timeouts	Explicit waits (20s), page load timeout 60s
UI locator changes	Broken scripts	POM centralises locators
Concurrent users	Session conflicts	Fresh browser per <code>@BeforeMethod</code>
Credential reset	Login failures	Credentials in <code>config.properties</code>

2.6 Observations

- OrangeHRM 5.x uses dynamic loads; `Thread.sleep` is unreliable
- Side menu uses text-based span labels (PIM, Admin)
- Login fields: `name="username"` and `name="password"`
- Invalid login shows alert: "Invalid credentials"
- Logout is under the top-right user dropdown

3 Step 3 - Manual Test Scenario Design (15 Marks)

Seventeen manual test scenarios were designed. Summary below; evidence screenshots included for high-priority cases.

ID	Title	Priority
TS-01	Valid login	High
TS-02	Invalid login	High
TS-03	Empty credentials	High
TS-04	Logout	High
TS-05	Dashboard loads after login	High
TS-06	Navigate to PIM	High
TS-07	Search employee by name	Medium
TS-08	Open Add Employee form	Medium
TS-09	Navigate to Admin / System Users	High
TS-10	Search system user	Medium
TS-11	Navigate to Leave	Medium
TS-12	Side menu collapse/expand	Low
TS-13	Forgot password link	Medium
TS-14	Direct dashboard URL when logged out	Medium
TS-15	Browser back after logout	Low
TS-16	Password field masks input	Medium
TS-17	Username only (password empty)	High

3.1 TS-01 - Valid Login

Steps: Open login URL → Enter Admin / admin123 → Click Login

Expected: Redirect to Dashboard; header shows "Dashboard"

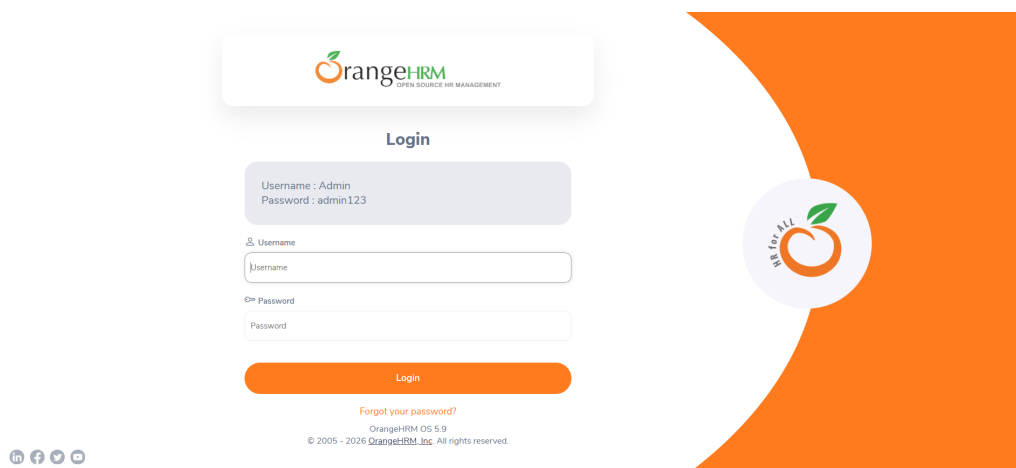


Figure 1: TS-01 - Login page

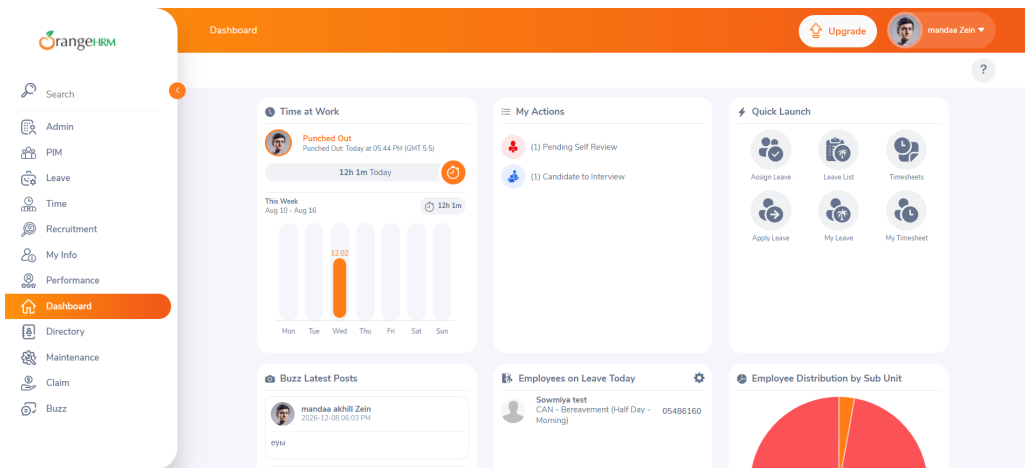


Figure 2: TS-01 - Dashboard after successful login

3.2 TS-02 - Invalid Login

Steps: Enter invalid username/password → Click Login

Expected: “Invalid credentials” error; remain on login page

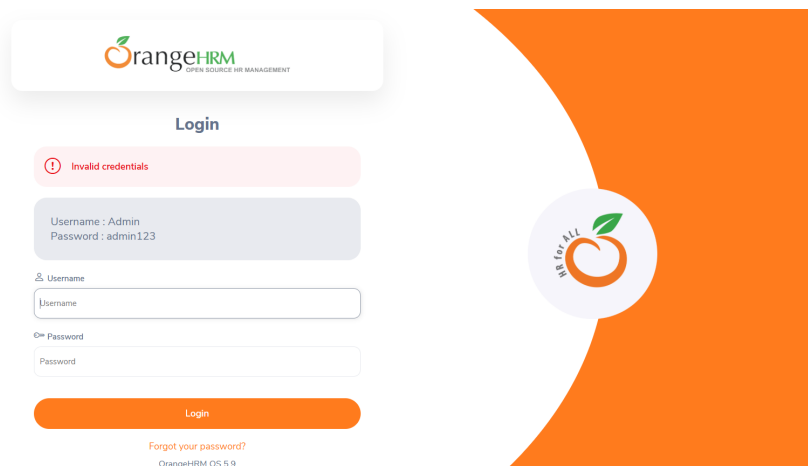


Figure 3: TS-02 - Invalid login error message

3.3 TS-05 - Dashboard Loads

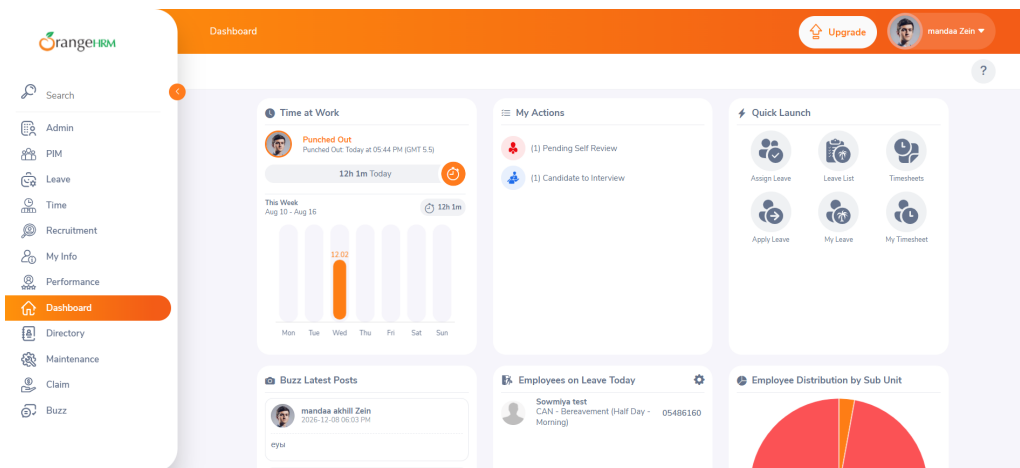


Figure 4: TS-05 - Dashboard widgets visible after login

3.4 TS-06 - Navigate to PIM

Expected: PIM page opens; URL contains pim

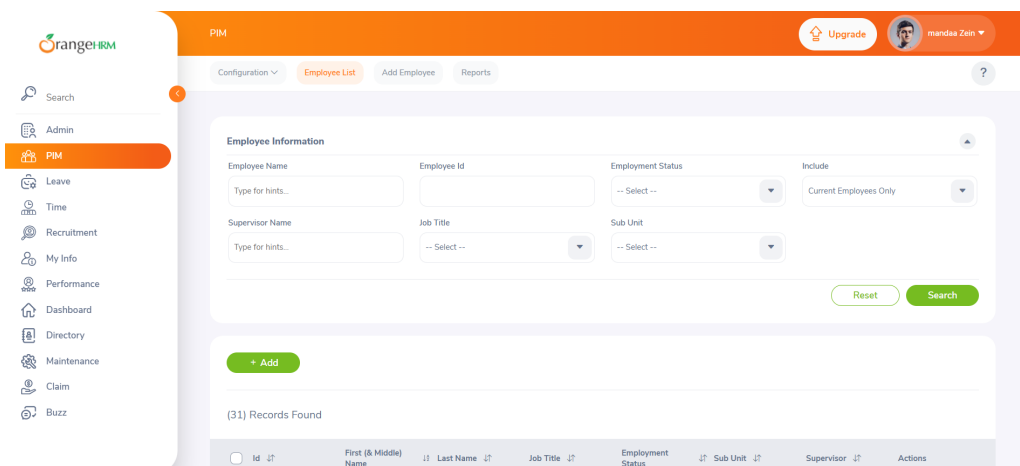


Figure 5: TS-06 - PIM module

3.5 TS-09 - Admin System Users

Expected: Admin module opens; “System Users” section visible

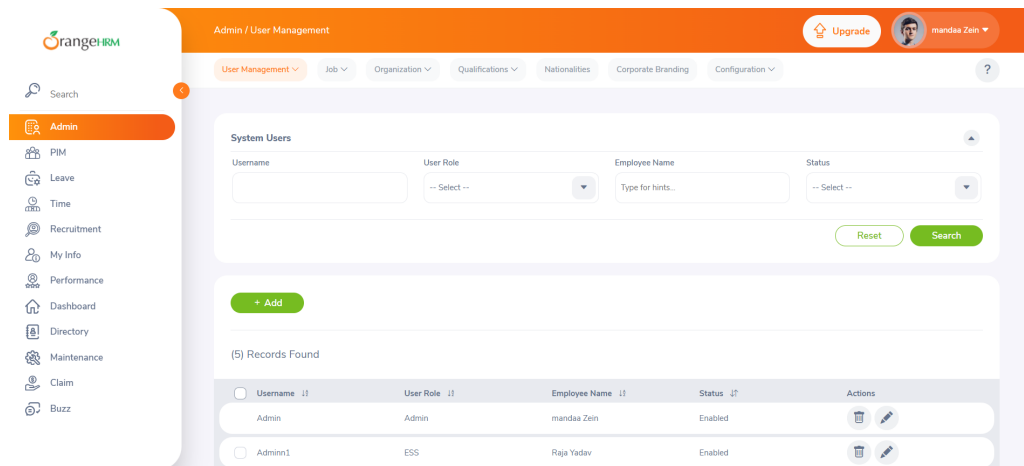


Figure 6: TS-09 - Admin System Users section

4 Step 4 - Automation Decision (10 Marks)

4.1 Selected for Automation (5 scenarios)

ID	Scenario	Why automate
TS-01	Valid login	Critical path; stable locators; strong regression ROI
TS-02	Invalid login	Negative auth; deterministic error message
TS-04	Logout	Security-related; clear URL/state check
TS-07	PIM search	Navigation + form + table coverage
TS-09	Admin users	Second major module; menu + search validation

4.2 Not Automated (examples)

- **TS-03** Empty credentials - overlaps TS-02; brittle UI validation
- **TS-08** Add Employee - creates data on shared demo; cleanup unreliable
- **TS-11** Leave flows - date pickers increase complexity
- **TS-12/16** Visual/UX checks - low automation value
- **TS-13** Forgot password - external/environment dependent

Decision rule: Prefer high priority, stable UI, low data side-effects, reusable smoke coverage.

5 Step 5 - Framework Design (15 Marks)

5.1 Technology Choices

- **Java 17** - industry-standard for Selenium
- **Maven** - dependency and build management
- **Selenium 4** - browser automation
- **TestNG** - test lifecycle, suite XML, assertions
- **Page Object Model** - separates UI from test logic

5.2 Package Structure

```
src/main/java/com/asqa/orangehrm/
    base/      BasePage (shared page helpers)
    pages/     LoginPage, DashboardPage, PIMPage, AdminPage
    utils/     ConfigReader, WaitUtils, ScreenshotUtils
src/test/java/com/asqa/orangehrm/
    base/      BaseTest (browser lifecycle)
    tests/     Five automated scenario classes
src/test/resources/
    config.properties, testng.xml
```

Package	Responsibility
base	Shared driver/page setup
pages	Locators and user actions (no TestNG asserts)
utils	Config, waits, screenshots
tests	Arrange-Act-Assert only
resources	Externalised config and suite XML

5.3 Design Principles

1. One browser session per @Test method
2. Explicit waits preferred over fixed sleeps
3. Screenshots captured on failure via @AfterMethod
4. Credentials stored in config.properties

6 Step 6 - Automation Development (20 Marks)

Test Class	Manual ID	Description
LoginValidTest	TS-01	Valid Admin login → Dashboard
LoginInvalidTest	TS-02	Invalid credentials → error
LogoutTest	TS-04	Logout → login URL
PIMSearchEmployeeTest	TS-07	Open PIM → search
AdminUserManagementNavigationTest	TS-09	Admin → System Users

Run command:

```
mvn clean test
```

All five automated tests pass successfully on Microsoft Edge.

7 Step 7 - Debugging Challenge (10 Marks)

7.1 Intentional Failure

The login button locator in LoginPage.java was changed from:

```
private final By loginButton = By.cssSelector("button[type='submit']");
```

to an incorrect selector:


```
private final By loginButton = By.cssSelector("button#login-btn-missing");
```

7.2 Observed Failure

- **Test:** LoginValidTest.validLogin_shouldOpenDashboard
- **Exception:** TimeoutException waiting for button#login-btn-missing
- **Stack trace:** WaitUtils.clickable → BasePage.click → LoginPage.clickLogin

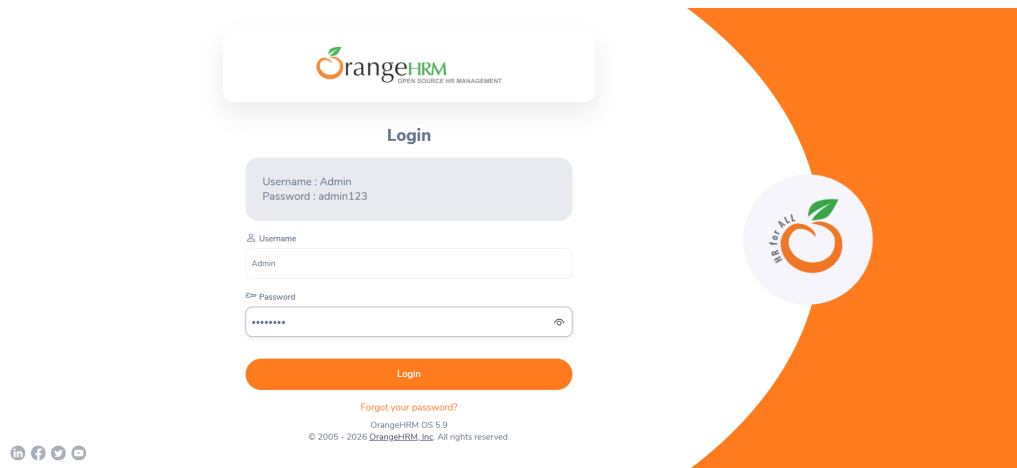


Figure 7: Debugging challenge - failure screenshot (login page visible, wrong locator)

7.3 Debugging Process

1. Read Surefire stack trace - pointed to login button wait timeout
2. Opened failure screenshot - page loaded but element not found
3. Inspected DOM - real button is `button[type='submit']`
4. Restored correct locator in `LoginPage` only (POM benefit)
5. Re-ran test - **BUILD SUCCESS**

7.4 Root Cause and Fix

Root cause: Invented CSS ID does not exist in OrangeHRM login DOM.

Fix: Restored `button[type='submit']`. Evidence logs in `docs/debugging/`.

8 Step 8 - Reflection Video (10 Marks)

A five-minute screen recording is submitted separately covering:

- Project structure walkthrough
- Running `mvn clean test`
- Page object and test class explanation
- Debugging challenge (before/after)
- Lessons learned (waits, POM, shared demo risks)

9 Step 9 - Viva Preparation (5 Marks)

Prepared to explain:

- Why Page Object Model separates locators from tests
- Why TestNG for browser lifecycle and suite control
- Explicit vs implicit wait strategy
- Why five scenarios were chosen for automation
- How `config.properties` improves maintainability

10 Step 10 - GitHub Repository (10 Marks)

Incremental commit history maintained at:

<https://github.com/SeniruR/QA-Automation-ASQA>

Commits follow logical progression: project setup → base utilities → page objects → tests → documentation → submission assets.

End of Report - R.P.S.R. Ranasinghe (22020782)